C1 - Fourier Transform

Part a)

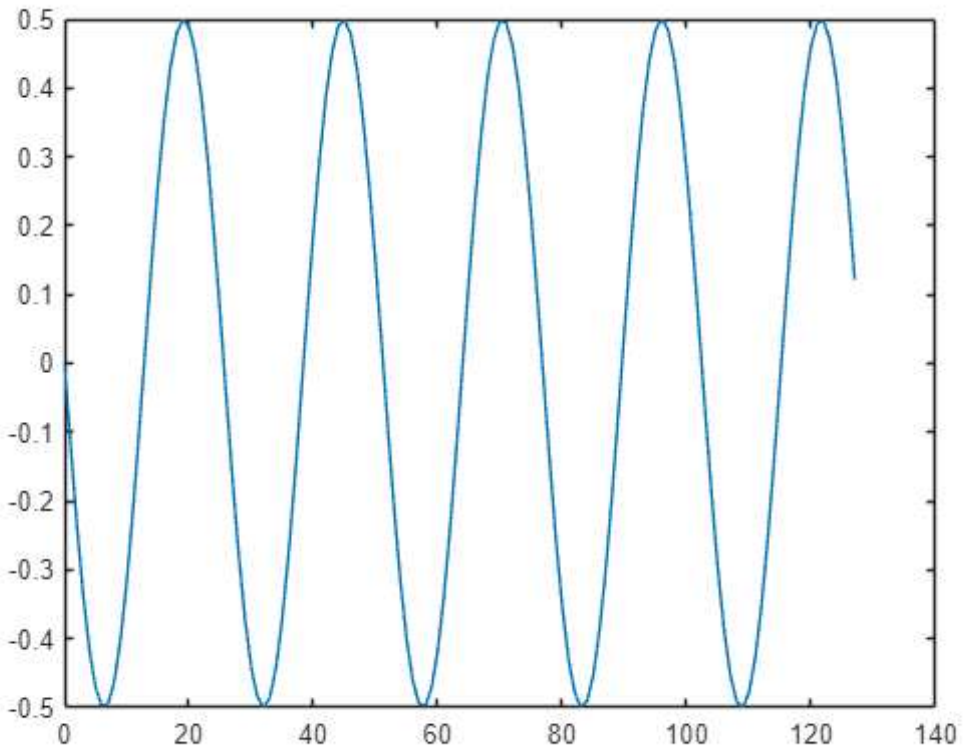
First, we define constants and the n-range

```
Nt = 128;
fs = 1000;
f0 = 5/128*fs;
a = 0.5;
phi = pi/2;
n = 0:Nt-1;
```

Then we define our anonymous function y(n) and plot it

```
y = @(n) (n<0).*(0) + (n>=0 & n<=Nt-1).*(a*cos(2*pi*f0*n/fs+phi)) + (n>Nt-1).*(0);

plot(n, y(n))
```



Part b)

Now we define constants for the transffourier function

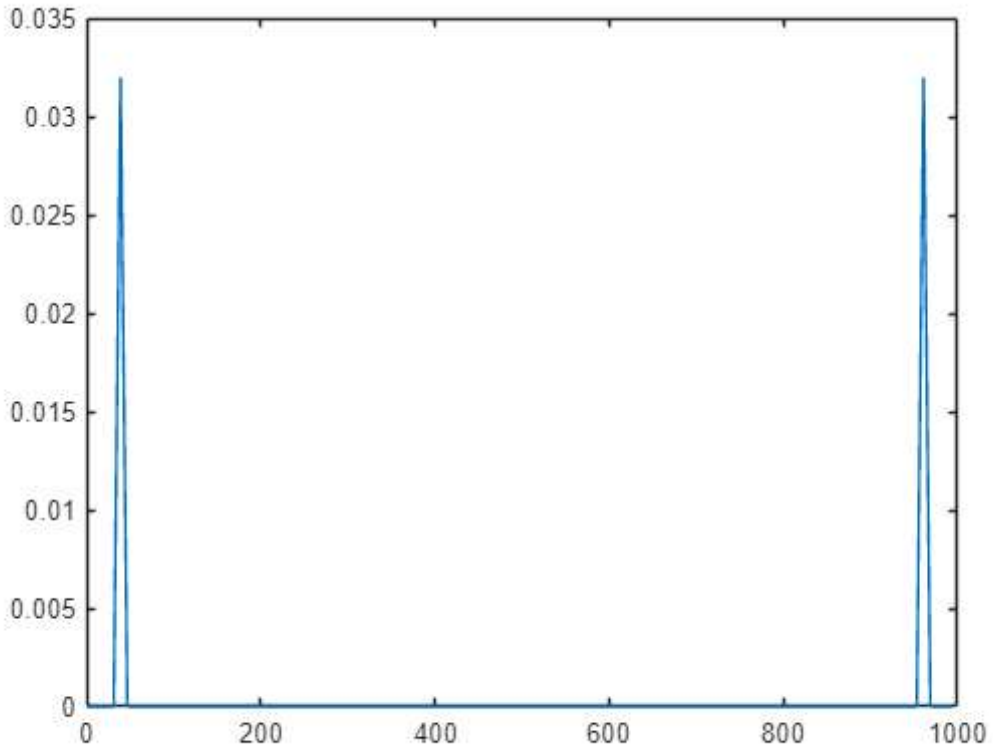
```
Nf = 128;
```

And we compute the discrete fourier transform

```
[f, tfx] = transffourier(y(n), Nf, fs);
```

and plot it in the frequency domain

```
plot(f, abs(tfx))
```



and to find the peaks we calculate:

```
[pks, locs] = findpeaks(abs(tfx), f, 'MinPeakHeight',0.1*max(abs(tfx)))
```

```
pks = 2×1
    0.0320
    0.0320
locs = 1×2
   39.0625  960.9375
```

to explain the positions of these peaks we compare the true and sampling frequencies based on the shannon formula

f_observed = min(k) |+/- f_actual + k*f_s|

```
fs
```

```
fs = 1000
```

```
f0
```

```
f0 = 39.0625
```

The minimum is reached for k=0 and we see

```
f_observed1 = +f0 + 0*fs
```

```
f_observed1 = 39.0625
```

which generates the peak at

```
locs(1)
```

```
ans = 39.0625
```

where the second peak at

```
locs(2)
```

```
ans = 960.9375
```

takes its value from k=1

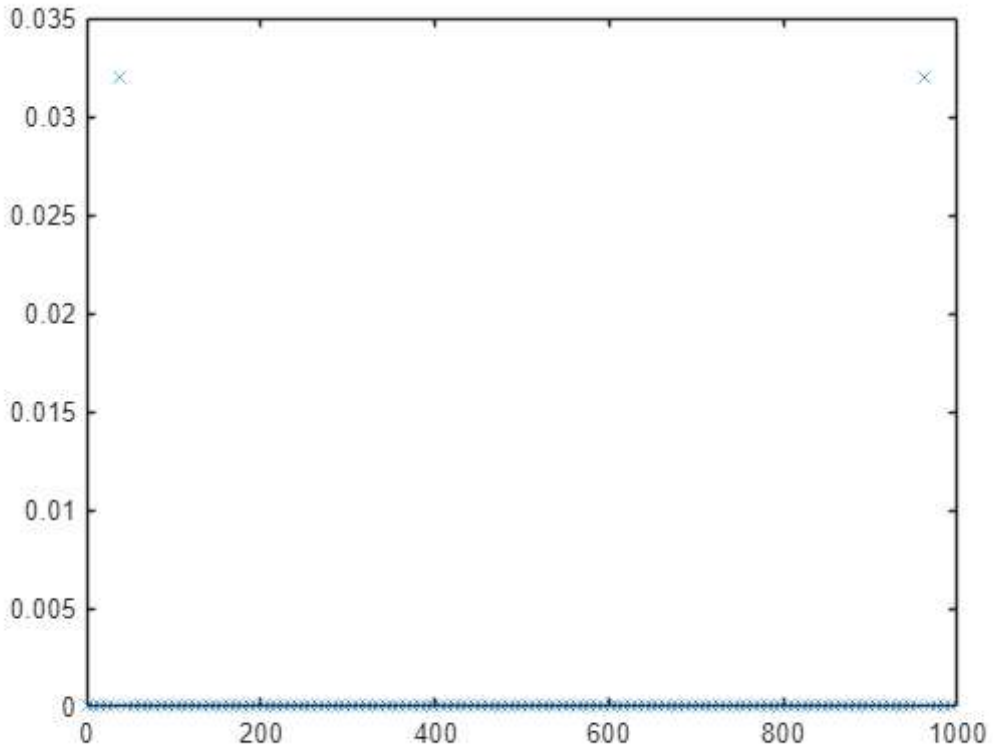
```
f_observed2 = -f0 + 1*fs
```

```
f_observed2 = 960.9375
```

Part c)

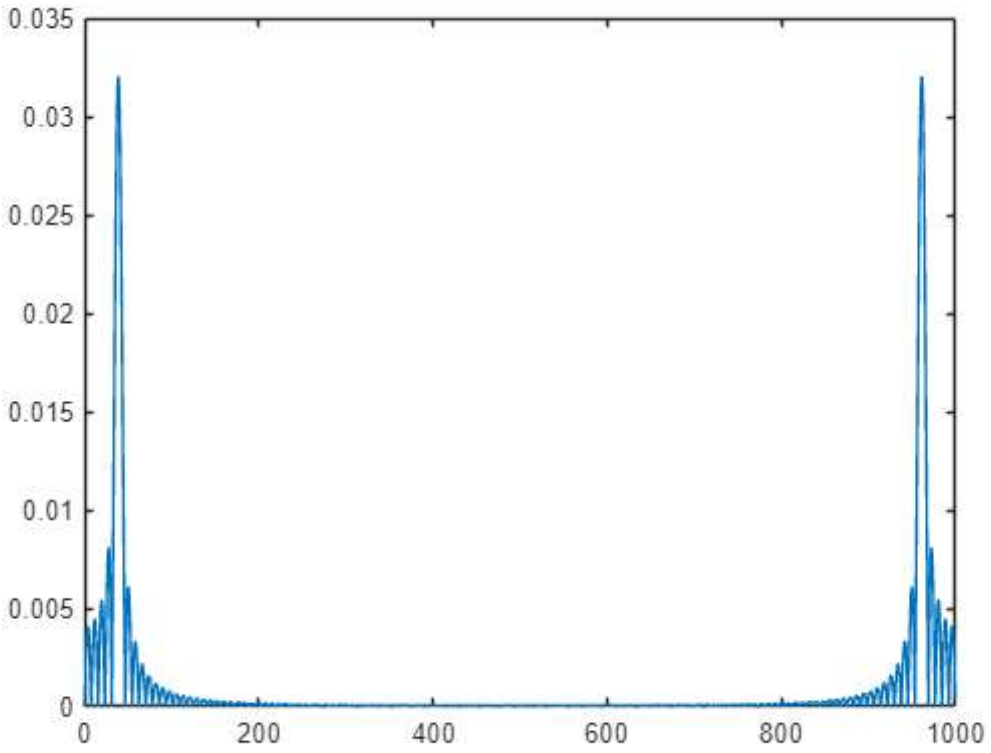
Instruction says to plot the magnitude spectrum with cross marker for Nf=128

```
plot(f, abs(tfx), 'x')
```



as well as the full line magnitude spectrum for Nf=4096

```
Nf2=4096;
[f2, tfx2] = transffourier(y(n), Nf2, fs);
plot(f2, abs(tfx2))
```



Part d)

The instructions say to redo all calculations but with f0=5.7/128*fs

```
f0 = 5.7/128 * fs
```

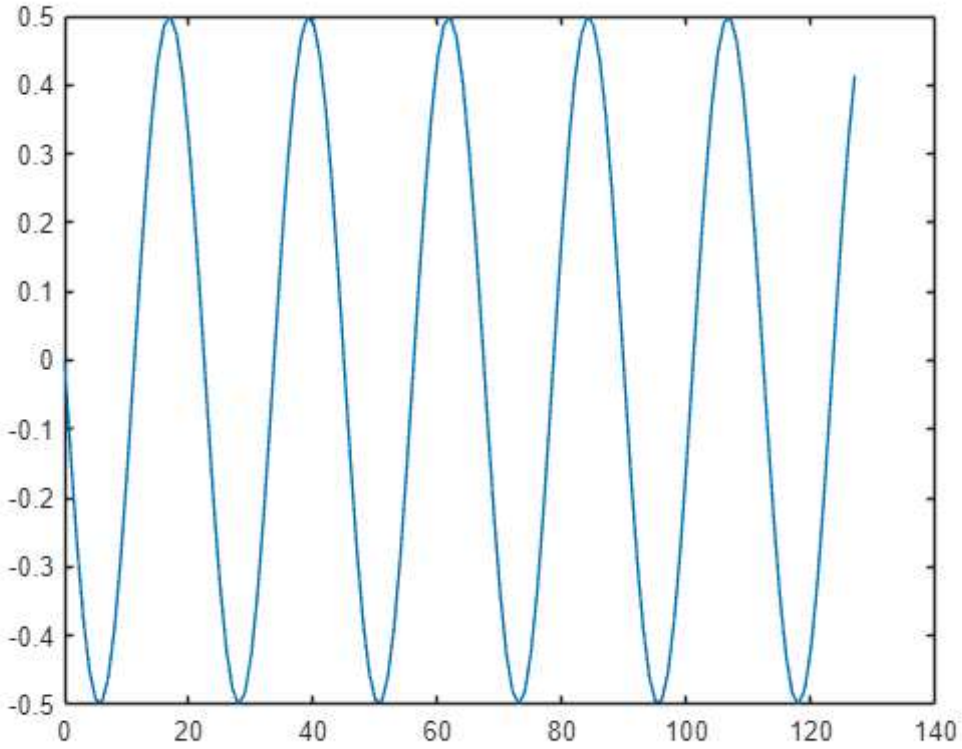
```
f0 = 44.5312
```

We need not initialise all constants and functions again, all that is required to start is

```
Nf = 128;
```

And we can proceed to part a)

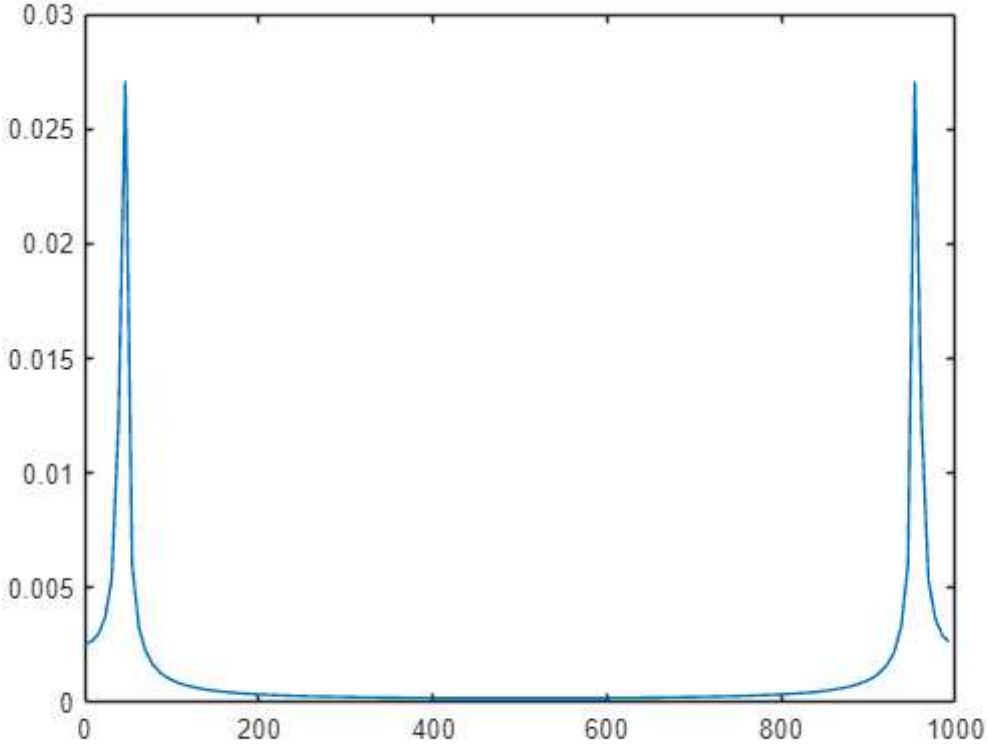
```
yd=@(n) (n<0).*(0) + (n>=0 & n<=Nt-1).*(a*cos(2*pi*f0*n/fs+phi)) + (n>Nt-1).*(0);
plot(n, yd(n))
```



then to part b)

the lines below in order are Compute, Plot, Peaks, f_observed from Shannon x2

```
[fd1, tfxd1] = transffourier(yd(n), Nf, fs);
plot(fd1, abs(tfxd1))
```




```
[pksd, locsd] = findpeaks(abs(tfxd1), fd1, 'MinPeakHeight',0.1*max(abs(tfxd1)))
```

```
pksd = 2×1
    0.0271
    0.0271
locsd = 1×2
   46.8750  953.1250
```

```
f_observed1 = +f0 + 0*fs
```

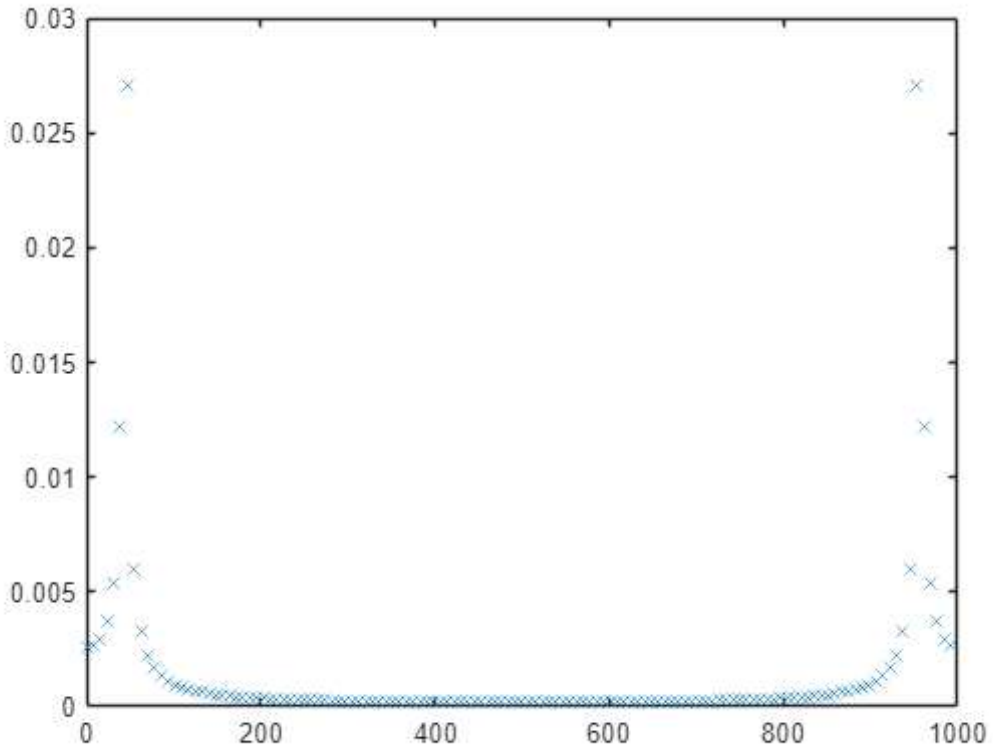
```
f_observed1 = 44.5312
```

```
f_observed2 = -f0 + 1*fs
```

```
f_observed2 = 955.4688
```

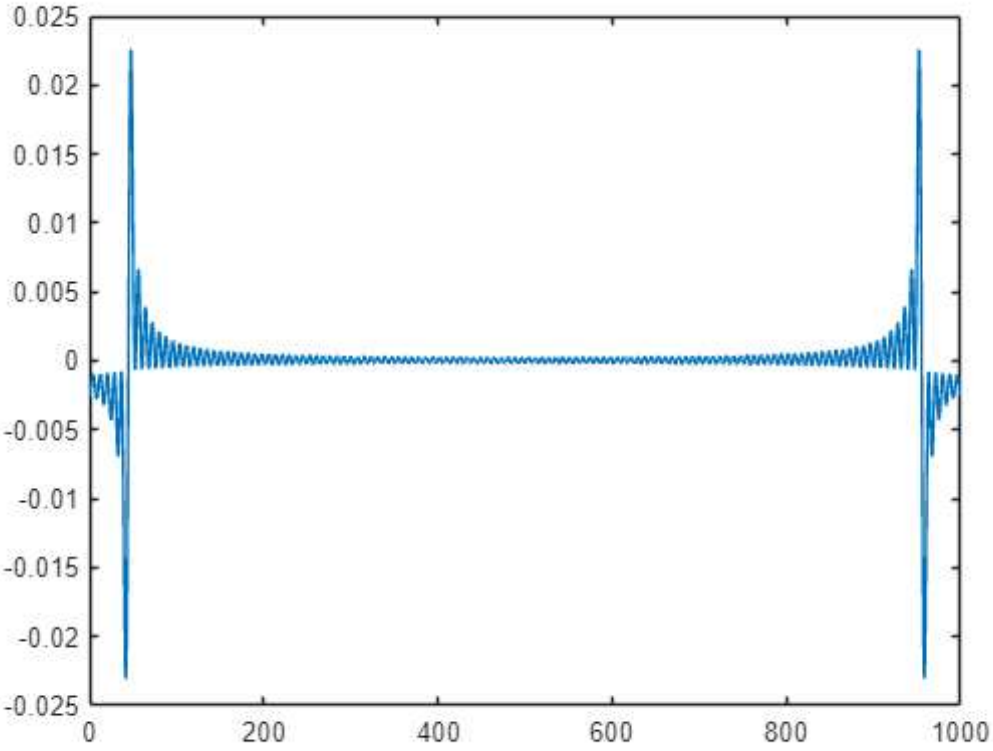
We can explain the divergence between locsd and f_observed by recalling Nf is a very small 128 as opposed to part c)

```
plot(fd1, abs(tfxd1), 'x')
```



And now for the full spectrum at 4096 frequencies

```
Nfd2=4096;
[fd2, tfxd2] = transffourier(yd(n), Nf2, fs);
plot(fd2, tfxd2)
```


0.025
0.02
0.015
0.01
0.005
0
-0.005
-0.01
-0.015
-0.02
-0.025
0
200
400
600
800
1000